DOCUMENT 03

UI/UX Specification and Design System

Screens, components, tokens, responsive and accessibility rules

Study Topic Extractor

Version: 1.0
Status: Initial Specification
Date: September 28, 2026

Turn long study material into a focused study roadmap.

Table of Contents

1. Design Principles 2

2. Design System: Tokens 3

2.1 Color Tokens 3

2.2 Typography Tokens 3

2.3 Spacing, Radius, Shadow, Border Tokens 4

3. Breakpoints and Responsive Strategy 5

4. Component Library 6

4.1 Priority Badge Spec 6

5. Global UX States 8

6. Page Specifications 9

6.1 Landing Page 9

6.2 Dashboard (Phase 2) 9

6.3 Analyze Document Page 9

6.4 Upload Page (Upload PDF tab and drop zone) 10

6.5 Processing State 10

6.6 Results Page 10

6.7 Topic Details Page / Drawer 11

6.8 Topic Relationship Graph (Phase 2) 11

6.9 Exam Focus Mode (Phase 2) 11

6.10 Document History (Phase 2) 12

6.11 Settings 12

6.12 Export Screen 12

7. Exam Focus Mode Content Rules (Phase 2) 13

8. Accessibility Specification 14

8.1 Accessible Error Messages 14

9. Acceptance Criteria 15

1. Design Principles

- **Readability over effects.** This is a study tool: text contrast, spacing and hierarchy come first.
- **Modern, minimal, student-friendly.** Gen-Z inspired but not childish: soft gradients, rounded cards, restrained glass effects.
- **Priority at a glance.** HIGH / MEDIUM / LOW is always conveyed by label + icon + color, never color alone.
- **Fast and calm.** Immediate feedback, honest progress states, no surprise modals.
- **Dark-first, light-complete.** Default follows system preference; both themes are first-class and use identical tokens.

2. Design System: Tokens

2.1 Color Tokens

Direction: dark futuristic educational SaaS with blue/purple/cyan accents. Primary dark background is `#0A0A0A`. Glassmorphism is limited to the navbar and modal backdrops (`backdrop-filter: blur(12px)`), never behind body text.

Token	Dark theme	Light theme	Usage
`--bg`	#0A0A0A	#FAFAFA	Page background
`--surface`	#111113	#FFFFFF	Sections, sidebar, navbar
`--card`	#16161A	#FFFFFF	Cards, inputs
`--border`	#2A2A31	#E4E4E7	Dividers, outlines
`--text`	#F4F4F5	#18181B	Primary text
`--text-muted`	#A1A1AA	#52525B	Secondary text (>= 4.5:1)
`--primary`	#4F46E5	#4F46E5	Buttons, active tabs (white text 6.3:1)
`--primary-soft`	#A5B4FC	#4338CA	Links, focus accents
`--secondary`	#22D3EE	#0E7490	Highlights, LOW priority
`--accent-purple`	#A78BFA	#6D28D9	Gradients, hierarchy level 2
`--success`	#4ADE80	#15803D	Success states
`--warning`	#FBBF24	#B45309	Warnings, MEDIUM priority
`--error`	#F87171	#B91C1C	Errors, HIGH priority
`--priority-high`	#F87171	#B91C1C	HIGH badge, bar
`--priority-medium`	#FBBF24	#B45309	MEDIUM badge, bar
`--priority-low`	#22D3EE	#0E7490	LOW badge, bar
`--focus-ring`	#A5B4FC	#4338CA	2 px outline, 2 px offset

Gradient: `--gradient-brand: linear-gradient(135deg, #4F46E5 0%, #7C3AED 50%, #06B6D4 100%)` used only in the hero, logo and primary CTA hover glow.

2.2 Typography Tokens

Font family: **Inter** (variable) for UI and body, **JetBrains Mono** for code and numbers in tables, both self-hosted via `next/font`. Fallback: `system-ui, -apple-system, Segoe UI, Roboto, sans-serif`.

Token	Desktop size / line height	Mobile size / line height	Weight
display	48 / 56 px	34 / 40 px	700
h1	36 / 44 px	28 / 36 px	700
h2	28 / 36 px	22 / 30 px	600
h3	22 / 30 px	20 / 28 px	600
h4	18 / 26 px	17 / 26 px	600
body	16 / 26 px	16 / 26 px	400
body-sm	14 / 22 px	14 / 22 px	400 / 500

Token	Desktop size / line height	Mobile size / line height	Weight
caption	12 / 18 px	12 / 18 px	500
mono	13 / 20 px	13 / 20 px	400

Minimum body size is 16 px; users may scale text to 200% without loss of content.

2.3 Spacing, Radius, Shadow, Border Tokens

Category	Tokens
Spacing (4 px base)	`space-0`=0, 1=4, 2=8, 3=12, 4=16, 5=20, 6=24, 8=32, 10=40, 12=48, 16=64, 20=80, 24=96 px
Radius	`sm`=6, `md`=10, `lg`=14, `xl`=20, `full`=9999 px. Cards use `lg`; buttons and inputs use `md`; badges use `full`.
Shadow (light)	`sm`: 0 1px 2px rgba(0,0,0,.06); `md`: 0 4px 12px rgba(0,0,0,.08); `lg`: 0 12px 32px rgba(0,0,0,.12)
Shadow (dark)	`sm`: 0 1px 2px rgba(0,0,0,.5); `md`: 0 4px 16px rgba(0,0,0,.5); glow: 0 0 0 1px rgba(99,102,241,.4), 0 8px 24px rgba(79,70,229,.25)
Border	1 px solid `--border`; emphasis 1 px `--primary-soft`; focus 2 px `--focus-ring`
Motion	`fast`=120 ms, `base`=200 ms, `slow`=320 ms; easing cubic-bezier(.2,.8,.2,1). Disabled under `prefers-reduced-motion`.
Z-index	base 0, sticky 10, dropdown 20, drawer 30, modal 40, toast 50

3. Breakpoints and Responsive Strategy

Name	Range	Layout notes
Mobile	< 640 px	Single column; bottom sheet drawers; tabs scroll horizontally; 16 px page padding
Tablet	640 - 1023 px	2-column grids; collapsible sidebar; 24 px padding
Laptop	1024 - 1279 px	Results in 2/3 + 1/3 layout; top navigation; 32 px padding
Desktop	1280 - 1535 px	Max content width 1200 px; persistent side panel
Large desktop	>= 1536 px	Max width 1360 px; three-pane results (list, detail, chart)

Mobile-first CSS. Touch targets are at least 44 x 44 px. Horizontal scrolling of the page body is never allowed; wide tables scroll within their own container.

4. Component Library

Component	Specification	States
Navbar	64 px height, glass surface, logo left, links center, theme toggle and account right; collapses to icon menu on mobile	default, scrolled (border), menu open
Sidebar (P2)	240 px, collapsible to 72 px icons; bottom tab bar on mobile	expanded, collapsed, active item
Button	Variants primary, secondary, ghost, danger; sizes sm 32, md 40, lg 48 px	default, hover, focus, active, disabled, loading (spinner + aria-busy)
Input / Textarea	40 px height, label above, helper and error text below	default, focus, filled, error, disabled
Text editor	Textarea with monospace-off body font, live counter `n / 150,000`, clear and sample buttons	empty, valid, too short, over limit
File uploader	Dashed drop zone, keyboard-activatable, file chip after selection	idle, drag-over, uploading, ready, error
Card	`--card`, radius lg, padding 20, border, hover lift 2 px	default, hover, selected, loading skeleton
Badge	Pill; priority badges show icon + text (HIGH, MEDIUM, LOW)	3 priorities, neutral, info
Progress bar	8 px, rounded, `role=progressbar` with `aria-valuenow`	determinate, indeterminate
Score bar	Horizontal bar 0-100 with numeric label and priority color	3 priorities
Chart	Plotly: score distribution bar, priority donut; themed via tokens; text alternative table	loading, empty, error
Topic card	Rank, name, score, priority badge, 2-line summary, up to 4 keyword chips, expand chevron	collapsed, expanded, selected, reviewed (P2)
Keyword chip	Pill, 28 px, border, optional count	default, hover, selected (filter)
Tabs	`role=tablist`, arrow-key navigation, underline indicator	default, active, disabled
Dropdown / Select	Native select on mobile; custom listbox on desktop	closed, open, disabled
Modal	Centered, max 560 px, focus trap, Escape closes, backdrop blur	open, closing
Toast	Bottom-right (top on mobile), auto-dismiss 5 s, `role=status`	success, info, warning, error
Tooltip	Shows on hover and focus, 300 ms delay, max 240 px	hidden, visible
Empty state	Illustration (simple SVG), title, one-line help, primary action	no results, no documents, no topics
Loading state	Skeleton blocks shaped like final content; spinner only inside buttons	-
Error state	Icon, human-readable message from error catalog, Retry button, request id in small text	recoverable, fatal

4.1 Priority Badge Spec

Priority	Score	Icon + label	Color token	Guidance text
HIGH	80 - 100	Filled circle + HIGH	`--priority-high`	Study first

Priority	Score	Icon + label	Color token	Guidance text
MEDIUM	50 - 79	Half circle + MEDIUM	`--priority-medium`	Review next
LOW	0 - 49	Ring + LOW	`--priority-low`	Supporting

5. Global UX States

State	Behavior
Loading	Skeletons for lists and cards. Stage-based progress for analysis. Buttons show spinner and are disabled while busy.
Empty	Explains why it is empty and offers the next action. Example: no topics found suggests providing more text with clear headings.
Error	Uses catalog messages (see Technical Requirements). Never shows stack traces. Offers Retry and, where useful, guidance.
Warning banner	Non-blocking notice (for example LOW_EXTRACTION_QUALITY or low language confidence), dismissible.
Offline	Toast: You appear to be offline. Retry when connected.
Long operation	After 15 s show: Large documents can take longer. Phase 2 offers to continue in the background.

6. Page Specifications

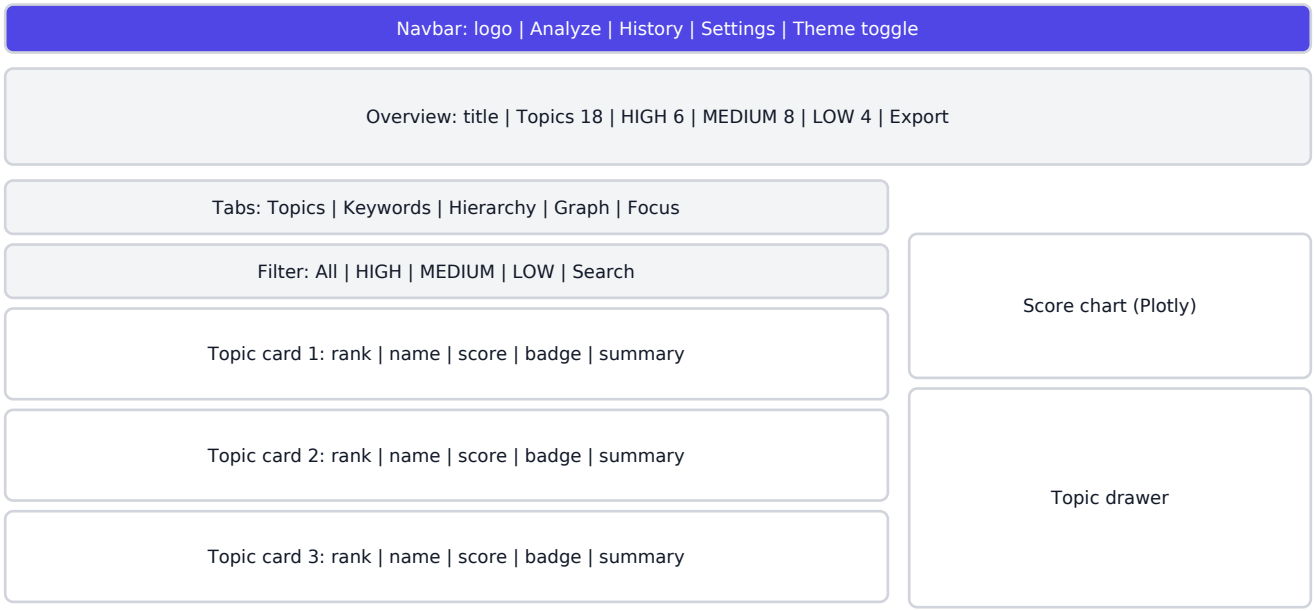


Figure 1. Results page wireframe (laptop)

6.1 Landing Page

Aspect	Specification
Purpose	Explain the value proposition and route users to Analyze.
Layout	Single column hero, three-step explainer, feature grid, sample output preview, footer.
Components	Navbar, Hero, CTA buttons, Step cards, Sample topic card, Footer.
User actions	Click Get Started / Analyze a document; toggle theme; open Settings.
States	Default; reduced-motion variant. No loading state (static).
Responsive behavior	Mobile: stacked; tablet: 2-column feature grid; laptop+: hero split with preview card.

6.2 Dashboard (Phase 2)

Aspect	Specification
Purpose	Personal overview: recent documents, study streak, weak areas.
Layout	Header with greeting, stat cards row, recent documents list, quick-start card.
Components	Stat cards, Document rows, Progress bars, Empty state.
User actions	Open a document; start new analysis; delete a document.
States	Loading skeleton; empty (no documents); error with retry.
Responsive behavior	Stat cards 1 / 2 / 4 columns; sidebar collapses to bottom tab bar on mobile.

6.3 Analyze Document Page

Aspect	Specification
Purpose	Primary input screen: paste text or upload PDF, then start analysis.
Layout	Centered card (max 880 px) with tabs Paste Text and Upload PDF, title field, Extract Topics button.
Components	Tabs, Text editor with counter, File uploader, Title input, Primary button, Helper text on limits and privacy.
User actions	Paste or type; upload; clear; press Extract Topics; try sample text.
States	Idle; text too short (button disabled + hint); over limit (error text); uploading; validation error; processing (see Processing State).
Responsive behavior	Full-width card on mobile; editor min height 240 px mobile, 360 px desktop.

6.4 Upload Page (Upload PDF tab and drop zone)

Aspect	Specification
Purpose	Accept a PDF via drag-and-drop or file picker with immediate validation.
Layout	Drop zone with dashed border, file summary row after selection, extraction preview and quality note.
Components	File uploader, File chip (name, size, pages), Progress bar, Quality alert, Remove button.
User actions	Select file; drag and drop; remove file; view extracted text preview; proceed to analyze.
States	Idle; drag-over; uploading with progress; extracted (preview + quality); invalid type; too large; encrypted; no readable text.
Responsive behavior	Drop zone becomes a tap target on touch devices; preview collapses to accordion.

6.5 Processing State

Aspect	Specification
Purpose	Show progress during upload extraction and analysis without blocking the UI.
Layout	Full-card overlay in place of the editor with stage list and indeterminate or stage-based progress.
Components	Progress bar, Stage list (Extracting, Cleaning, Finding topics, Ranking, Writing summaries), Cancel button.
User actions	Cancel analysis (aborts request).
States	Running; slow (> 15 s hint: large documents take longer); failed (error state with retry); completed (auto-navigate).
Responsive behavior	Same layout at all sizes; stage list becomes a single status line on mobile. Announced through aria-live polite.

6.6 Results Page

Aspect	Specification
Purpose	Present the study roadmap: overview, ranked topics, keywords, side chart.
Layout	Overview strip, tab bar, filter row, topic card list (main), side panel with score distribution and keywords.

Aspect	Specification
Components	Overview cards, Tabs, Filter chips, Topic cards, Priority badges, Keyword chips, Plotly chart, Export button.
User actions	Filter by priority; search; open topic detail; switch tabs; export; start new analysis.
States	Loading skeleton; results; no topics found (empty state with tips); partial (warning banner); error.
Responsive behavior	Mobile: single column, side panel becomes a Summary tab. Laptop: 2/3 + 1/3. Large desktop: adds persistent detail panel.

6.7 Topic Details Page / Drawer

Aspect	Specification
Purpose	Explain one topic: score breakdown, summary, keywords, source sentences, relations, questions.
Layout	MVP: right-side drawer (bottom sheet on mobile). Phase 2: full page `/results/[id]/topics/[topicId]`.
Components	Header with priority badge, Score breakdown bars (tfidf, frequency, position, semantic), Summary, Source sentences, Related topics, Possible questions (P2).
User actions	Close; next/previous topic; copy summary; mark reviewed (P2).
States	Default; no related topics; loading (P2).
Responsive behavior	Drawer 420 px on desktop; bottom sheet 85% height on mobile; focus trapped, Escape closes.

6.8 Topic Relationship Graph (Phase 2)

Aspect	Specification
Purpose	Visualize topic hierarchy and semantic links.
Layout	Canvas graph with controls (zoom, fit, layout), legend, side detail on selection.
Components	Cytoscape canvas, Legend, Edge-type toggles, Zoom controls, Accessible list alternative.
User actions	Pan, zoom, select node, toggle edge types, switch to list view.
States	Loading; empty (fewer than 3 topics); large graph (cap 40 nodes); error.
Responsive behavior	Mobile: graph fullscreen with bottom controls; list view default on screens < 480 px.

6.9 Exam Focus Mode (Phase 2)

Aspect	Specification
Purpose	Revision-oriented views: Quick Revision, Normal Study, Deep Study.
Layout	Mode selector (segmented control) above a stacked content area.
Components	Segmented control, Topic list, Definition cards, Question list, Checklist with progress.
User actions	Switch mode; check off topics; reveal answers hint (extractive source); print view.
States	Per-mode content; loading questions; none available.
Responsive behavior	Segmented control becomes a dropdown on mobile; two columns in Deep Study on desktop.

6.10 Document History (Phase 2)

Aspect	Specification
Purpose	List, search and reopen saved analyses.
Layout	Table (desktop) or card list (mobile) with title, date, topic count, top priority topics.
Components	Search input, Sort dropdown, Rows, Overflow menu, Delete confirmation modal.
User actions	Open, rename, delete, export.
States	Loading; empty; error; deleting.
Responsive behavior	Table with sticky header on desktop; cards on mobile.

6.11 Settings

Aspect	Specification
Purpose	Theme, default study mode, scoring options, privacy controls.
Layout	Sections in a single column: Appearance, Analysis defaults, Data and privacy, Account (P2).
Components	Toggle, Radio group, Sliders for weights (Advanced), Number input for max topics, Danger zone.
User actions	Change theme; adjust max topics; edit weights (must sum to 1 with live validation); reset defaults; delete data (P2).
States	Saved toast; validation error; reset confirmation modal.
Responsive behavior	Single column at all sizes; max width 720 px.

6.12 Export Screen

Aspect	Specification
Purpose	Choose format and download results.
Layout	Modal (desktop) or full-screen sheet (mobile) with format cards and options.
Components	Format cards (PDF, CSV, TXT, MD, JSON), Options (include summaries, keywords, priority filter), Download button, Toast.
User actions	Select format; toggle options; download; cancel.
States	Idle; generating (spinner); success toast; failure toast with retry.
Responsive behavior	Cards 1 / 2 / 3 columns.

7. Exam Focus Mode Content Rules (Phase 2)

Mode	Shows	Hides	Use case
Quick Revision	Top topics (HIGH), important keywords, one-line definitions, key concepts checklist	Explanations, relationships, questions	Last-minute revision
Normal Study	Topics, subtopics, short explanations, keywords, priority	Detailed explanations, questions	Regular study
Deep Study	Topics, subtopics, definitions, relationships, concepts, detailed explanations (2 source sentences), possible questions	-	Exam preparation

8. Accessibility Specification

- **Standard:** WCAG 2.2 AA.
- **Keyboard:** all functionality reachable by keyboard; logical tab order; visible 2 px focus ring; arrow keys inside tabs and segmented controls; Escape closes drawers and modals.
- **Contrast:** text >= 4.5:1, large text and UI components >= 3:1, verified for both themes.
- **Priority not by color alone:** label + icon + position in list.
- **Screen readers:** semantic landmarks (`header`, `nav`, `main`, `footer`), headings in order, labels on all inputs, `aria-live=polite` for processing progress, `aria-busy` on loading regions.
- **Forms:** errors linked with `aria-describedby`; error summary focus after failed submit.
- **Graph:** provide equivalent nested list; nodes focusable; selection announced.
- **Motion:** respect `prefers-reduced-motion`; no auto-playing animation longer than 5 s.
- **Charts:** each chart has a data table alternative and text summary.

8.1 Accessible Error Messages

Situation	Message	Announcement
Invalid PDF	Please upload a valid PDF file.	role=alert
No text	We couldn't find readable text in this document.	role=alert
Too large	This document is larger than the supported processing limit.	role=alert
Processing failure	We couldn't analyze this document. Please try again.	role=alert

9. Acceptance Criteria

1. All twelve pages implement listed states and responsive behavior (Phase 2 pages when in scope).
2. Light and dark themes use identical tokens with values above; no hard-coded colors in components.
3. Automated axe scan shows zero critical or serious issues on Analyze, Results and Settings.
4. Keyboard-only user can complete paste, analyze, filter, open topic, export.
5. Lighthouse accessibility score >= 95 on Landing and Results.
6. No horizontal page scroll at 320 px width.